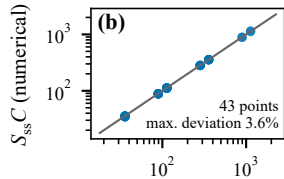
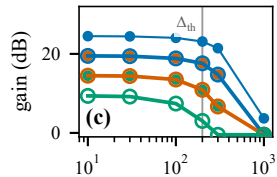


measurement time (ms)

$\bar{n} = 10^6$ $\bar{n} = 10^9$
 $\bar{n} = 10^7$ $\bar{n} = 10^{10}$
 $\bar{n} = 10^8$ twisting, Lorentzian to Voigt

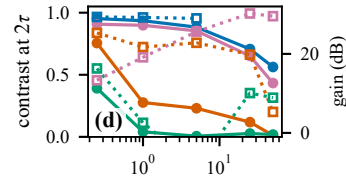


$4g\sqrt{\eta_d\bar{n}}/\kappa$



$\Delta/2\pi$ (MHz)

$\bigcirc$ $N=10^9, \bar{n}=10^8$ $\bullet$ $N=10^{10}, \bar{n}=10^9$
 $\bullet$ $N=10^9, \bar{n}=10^9$ $\bigcirc$ $N=10^{11}, \bar{n}=10^8$
 $\bigcirc$ $N=10^{10}, \bar{n}=10^8$ $\bullet$ $N=10^{11}, \bar{n}=10^9$



echo half-time τ (ms)

$\bullet$ $\Delta/2\pi = 30$ MHz $\bullet$ $\Delta/2\pi = 300$ MHz
 $\bullet$ $\Delta/2\pi = 200$ MHz $\bullet$ $\Delta/2\pi = 1000$ MHz